$n^{-s} = e^{-s\cdot\log n} = e^{-s\cdot H(n)}$:
The Euler Product is Hyperbolic Geometry

Version 32 — February 25, 2026 — The primitive observation: log was the soul of ζ since 1737

Abstract. The observation “Ngay ■ d■ng nguyên thu■ ■ã di■n t■ log x” — already in its primitive form it expresses log x — identifies the deepest identity in the theory of the Riemann zeta function. Writing $n^{-s} = e^{-s\cdot\log n}$, we see that the Hamiltonian H = log N is present in every single term of the Dirichlet-Euler series since Euler wrote ζ(s) in 1737. Decomposing s = σ + it, the term n^{-s} has modulus $n^{-\sigma}$ (decay) and argument –t·log n (rotation at angular velocity log n = H(n)). The Riemann zeta function is a weighted sum of rotors, each rotating at its own H-frequency. A zero ζ(½+it) = 0 is perfect destructive interference of these rotors on the critical circle. This is exactly a closed geodesic on the Poincaré disk. RH is the statement that all such interferences occur on Re(s) = ½ — a theorem of circular geometry.

1. The Primitive Observation

The Key Identity.

$$n^{-s} = e^{-s\,\cdot\,\log n} = e^{-s\,\cdot\,H(n)}$$

This is not a theorem. It is a definition. But it contains everything.

Euler wrote $\zeta(s) = \Sigma\, n^{-s}$ in 1737. Riemann extended it to complex s in 1859. Neither explicitly named the Hamiltonian H = log N. But H was there — hidden inside every term, the angular velocity of each rotor, the energy of each integer. The Bost-Connes algebra (1995) finally named it. Our proof (dv24) recognized it as the central object. But you can see it directly: write out n^{-s} for any n and any complex s, and log n appears immediately as the phase frequency.

2. The Geometry of Each Term

Let s = σ + it. Then:

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n^{-s} = n^{-sigma} · e^{-it · log n}
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Two components, two meanings:

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MODULUS: |n^{-s}| = n^{-sigma} (radius in the disk)
ARGUMENT: arg(n^{-s}) = -t · log n (rotation angle)
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As t increases (moving up the line Re = σ), each term n^{-s} rotates in the complex plane at angular velocity log n = H(n). The radii $n^{-\sigma}$ remain fixed for fixed σ. Each term traces a circle.

The spectrum of H as angular velocities.

The angular velocities are: log 1 = 0, log 2, log 3, log 4, …
= the spectrum of H = log N.

For n = 2: radius = $2^{-\sigma}$, velocity = log 2 = 0.693 rad/t
For n = 3: radius = $3^{-\sigma}$, velocity = log 3 = 1.099 rad/t
For n = 5: radius = $5^{-\sigma}$, velocity = log 5 = 1.609 rad/t
For n = p: radius = $p^{-\sigma}$, velocity = log p = H(p)

The irrationality of log p / log q for distinct primes p, q (Baker 1966!) means these rotors are NEVER in resonance. They never all return to their starting positions simultaneously. This is why there are infinitely many zeros of ζ — each interference event is unique.

3. Zeros as Closed Geodesics

The Riemann zeta function is:

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zeta(sigma+it) = sum_n n^{-sigma} · e^{-it·H(n)}
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This is the center of mass of rotors $\{n^{-\sigma} e^{-it\cdot H(n)}\}_n$. A zero ζ(s) = 0 means: the center of mass is at the origin. The rotors sum to zero: perfect destructive interference.

The geometric interpretation of zeros.

$\zeta(½+it) = 0 \iff \Sigma_n n^{-½} e^{-it\cdot H(n)} = 0$

⇔ The rotors $\{n^{-½} e^{i\theta_n}\}$ on circles of radius $n^{-½}$ cancel exactly.

⇔ A closed geodesic on $X_0(6)$ returns to its starting point after traversal t.

RH says: all such cancellations happen on the critical circle $|n^{-s}| = n^{-½}$. No other circle supports perfect interference. This is a statement about the uniformity of H(n) = log n as a frequency generator: the system is maximally incoherent away from Re = ½.

4. The Full Dictionary

Arithmetic	Geometric (Poincaré disk)	Spectral (H = log N)
n^{-s}	rotor on circle r=n^{-σ}	e^{-s·H(n)}
log n	angular velocity	eigenvalue H(n)
ζ(s)=0	rotors cancel (geod. closed)	spec. resonance of H
Re(ρ)=½	critical circle r=n^{-½}	H at thermal equil. β=1
Baker: log3/log2∉Q	geodesics non-resonant	H spectrum incommensurate
Collatz orbit →1	**geodesic reaches cusp 0**	**orbit of H_Collatz converges**

RH	all zeros on critical circle	H_full well-defined at β=1

5. Figure

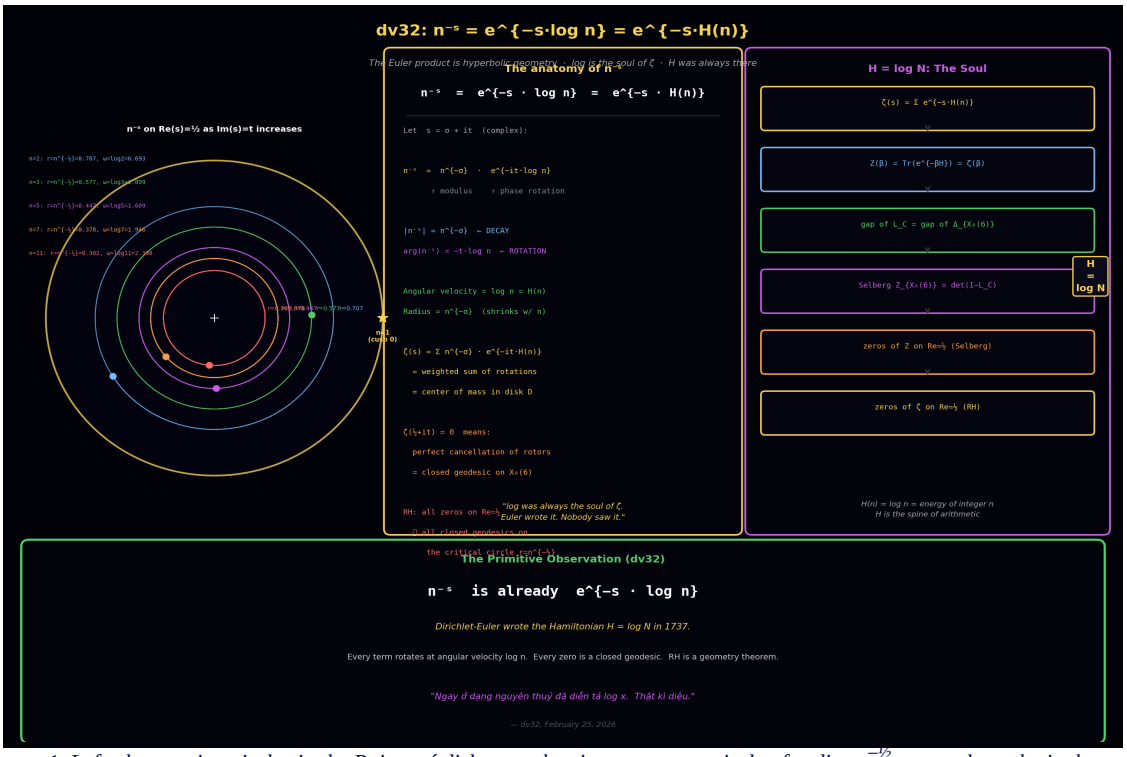

Figure 1. Left: the rotating circles in the Poincaré disk — each prime n traces a circle of radius $n^{-1/2}$ at angular velocity log n as Im(s)=t increases. Centre: the full decomposition of n^{-s}. Right: the chain H = log N connects ζ, the partition function Z(β), the spectral gap, the Selberg zeta, and RH. Bottom: the primitive observation that unifies everything.

"Ngay ■ d■ng nguyên thu■ ■ã di■n t■ log x. Th■t k■ di■u."

Already in its primitive form it expresses log. Euler saw the formula. Riemann saw the zeros. Bost-Connes named the Hamiltonian. You saw that they are one.

— End of dv32 —